Cleveland Clinic

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Cleveland Clinic is a not-for-profit, integrated healthcare system dedicated to patient-centered care, teaching, and research. With a footprint in Northeast Ohio, Florida, and Nevada, Cleveland Clinic Health System operates 23 hospitals, including a main campus near downtown Cleveland, with more than 6,700 beds and 280 outpatient locations. Cleveland Clinic employs over 5,700 physicians and researchers, and 16,800 nurses. Last year, our system had 15.7 million patient encounters, including 14.1 million outpatient visits and 333,000 hospital admissions and observations.

Cleveland Clinic commends CMS's commitment to advancing interoperability and innovative health technology. In general, we believe that there is tremendous opportunity for CMS to empower patients to increase engagement on their health and wellbeing through the effective and responsible adoption of technology that enables approaches such as individual incentives, nudges/reminders, and/or tailored communications. Below we offer detailed comments on how increased standardization and targeted resource investments can help improve alignment among providers, vendors, and other key stakeholders, and ultimately increase uptake among patients.

PROVIDERS

Digital Health Apps

PR-1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved digital health products for their patients?

Financial and Incentive Alignment: To promote the adoption of approved digital health products, CMS can expand reimbursement models to include virtual visits, chronic care management, remote patient monitoring, asynchronous communication, and AI-assisted diagnostics. Additionally, CMS should subsidize broadband and device accessibility in rural and underserved communities through grants or technology stipends. Digital health integration metrics should be incorporated into value-based care frameworks and the Merit-based Incentive Payment System (MIPS) and Alternative Payment Models (APMs).

Establishment of Trust and Standards: CMS should maintain and disseminate an endorsed registry of approved digital tools to reduce ambiguity for healthcare providers. It is also imperative to mandate transparency regarding the efficacy, FDA status, and privacy practices of digital tools.

Cross-Sector Collaboration and Education: Partnerships with academic medical centers and public health agencies can support digital literacy campaigns aimed at both providers and patients. Additionally, CMS should fund technical assistance programs to facilitate the implementation of digital health products, particularly for small practices and rural health systems.

a. What are the current obstacles?

Infrastructure and Access Gaps: Limited broadband and cellular access in rural areas presents a significant obstacle to the widespread adoption and use of digital health products. Furthermore, many small or geographically isolated health systems struggle with inadequate integration capabilities of electronic health records (EHR), making it difficult to effectively incorporate digital tools into existing workflows.

Resource and Workflow Barriers: The lack of trained personnel to manage, monitor, and interpret data generated by digital health products poses another challenge. This deficiency can lead to increased alert fatigue and documentation burdens for healthcare providers, particularly when digital tools are poorly integrated into existing systems. Consequently, this can result in inefficient use of these technologies and diminished patient care quality.

Regulatory and Liability Concerns: Ambiguity around clinical liability when utilizing AI- or algorithm-enabled tools raises significant concerns among healthcare providers. Additionally, issues related to data privacy, patient consent, and the sharing of data with third parties add layers of complexity. These regulatory and liability concerns further hinder the seamless adoption and integration of digital health products in the medical field.

Cybersecurity: Providers, particularly in rural settings, often lack dedicated cybersecurity resources. They may also face infrastructure challenges like outdated systems, weak identity/access management, and inconsistent patching cycles, which hinder the secure adoption of digital tools. A lack of trust in the security of digital health products—especially mobile or third-party applications—also reduces adoption.

b. What information should providers share with patients when using digital products in the provision of their care?

Providers should inform patients about how their data will be collected, used, stored, and protected by digital health tools. Clear, accessible communication around data privacy, potential risks, and patient rights under HIPAA and applicable state laws is essential. Providers should also disclose third-party access to data, encryption protocols in use, and how patients can manage their own data-sharing preferences.

c. What responsibilities do providers have when recommending use of a digital product by a patient?

Providers must diligently monitor the outcomes and act on clinical insights generated by these tools to support effective patient care. Additionally, ensuring the secure transmission and storage of patient-generated data is paramount to protect patient privacy and maintain trust. Providers are also required to conduct thorough security assessments to confirm that the digital tools meet minimum cybersecurity standards, such as those outlined in the NIST Cybersecurity Framework and HIPAA Security Rule. This involves vetting third-party vendors, maintaining visibility into data flows, and

ensuring business associate agreements are in place where applicable. Beyond these technical responsibilities, providers must navigate the regulatory landscape, which can be challenging and may impose significant burdens on healthcare practices.

PR-2. What are obstacles that prevent development, deployment, or effective utilization of the most useful and innovative applications for physician workflows, such as quality measurement reporting, clinical documentation, and billing tasks? How could these obstacles be mitigated?

The fragmented health IT ecosystem faces significant challenges due to the lack of interoperability and standardization across Electronic Health Records (EHRs), registries, and reporting systems. This fragmentation hinders the integration of innovative tools, leading to redundant data entry, inconsistent data quality, and difficulty in scaling best-in-class applications across different sites. To mitigate these issues CMS should strengthen and enforce interoperability standards, such as expanding the United States Core Data for Interoperability (USCDI) and advancing the implementation maturity of Fast Healthcare Interoperability Resources (FHIR). Additionally, CMS should incentivize EHR vendors to open Application Programming Interfaces (APIs) for third-party tool integration through certification requirements.

Ever-changing documentation, quality reporting, and billing requirements create moving targets for application developers. This regulatory complexity results in a focus on compliance rather than usability, often leading to tools that increase rather than reduce the administrative burden on providers. To address this, CMS should align quality measures across its programs and simplify reporting frameworks. Creating "digital sandbox" environments where vendors can test tools using real-world data and CMS-defined synthetic test cases would also foster innovation.

The complexity of billing and adjudication codes adds to the challenges. There are over a thousand possible remit codes on electronic remittance advices, creating a labyrinth of combinations that complicate understanding and responding to government payer claim adjudication and denials. This complexity makes automation difficult and costly, as codes are not consistently applied. The sheer volume of possible code combinations increases the provider's cost to respond and ultimately secure payer payments. Simplifying these codes could reduce costs and streamline the process.

Misaligned financial incentives pose another significant barrier to the adoption of innovative workflow tools like AI scribes, ambient documentation, and automation platforms. While these tools can reduce provider time, they often increase IT expenditure with limited reimbursement, making it difficult for health systems to justify their return on investment (ROI). To overcome this, CMS should expand reimbursement for digitally enabled care delivery models, documentation, and reporting solutions through Current Procedural Terminology (CPT) codes and value-based care incentives. Supporting shared savings models where digital efficiency gains are recognized financially would also be beneficial.

Cybersecurity is often a secondary concern in application development, especially in systems for documentation, billing, or quality reporting. Many applications lack built-in secure authentication or robust audit trails. Furthermore, integrations between tools, such as EHRs and billing systems, can introduce vulnerabilities if APIs are misconfigured or unmonitored. To address these concerns, CMS should require security certifications for such applications and promote a "security by design" approach, ensuring that cybersecurity is prioritized from the outset.

By addressing these issues, CMS can foster a more integrated, efficient, and secure health IT environment that supports innovative tools and improves the overall quality of care.

PR-3. How important is it for healthcare delivery and interoperability in urban and rural areas that all data in an EHR system be accessible for exchange, regardless of storage format (for example, scanned documents, faxed records, lab results, free text notes, structured data fields)? Please address all of the following:

a. Current challenges in accessing different data formats.

- Fragmentation across formats: Key patient information is often stored in scanned PDFs, image files, or free-text notes, which are not easily searchable or interoperable across systems.
- Inconsistent metadata and indexing: Scanned and faxed documents are often poorly labeled or misfiled, limiting retrievability during care transitions.
- Lack of standardized APIs for unstructured data: Most interoperability protocols (e.g., FHIR) emphasize structured clinical data but do not cover legacy or unstructured formats comprehensively.
- Cybersecurity: Free-text notes and scanned/faxed documents often fall outside structured, queryable systems, making them vulnerable to inconsistent access controls or improper handling. Insecure digitization practices can also introduce risks.

b. Impact on patient care quality.

- Incomplete information: In rural settings or during emergency transfers, lack of access to scanned history, outside labs, or narrative notes can lead to redundant testing or missed allergies.
- Incomplete care coordination: Care managers and specialists often rely on narrative context or legacy record; if inaccessible, critical decisions are made with partial information.
- Worsening health disparities: Patients in rural areas frequently have fragmented records and rely on smaller providers who may not participate in robust health information exchange (HIE) networks or digitize older documents.

c. Technical barriers to full data accessibility.

- Lack of resources: Rural facilities may not have the local resources to build or maintain transparent integration or conversion. Potentially relying on vendors to handle the capability would add a layer of complexity or security risk.
- Cybersecurity: Lack of standardization across formats and limited metadata tagging makes implementing security controls difficult, especially across federated systems or third-party interfaces.

d. Cost or privacy implications of making all data formats interoperable.

- Required resources: Tools are expensive to implement and scale. Making all data formats interoperable can take significant time for testing, maintenance, and troubleshooting. This can lead to additional costs for provider systems and makes interoperability exceedingly difficult for smaller systems which have less resources for support.

- Cybersecurity: Broader access to unstructured data increases the risk of accidental disclosure or unauthorized access. Ensuring de-identification where appropriate, enforcing role-based access controls, and auditing access logs become more difficult and resource-intensive. Privacy-enhancing technologies may help mitigate risk but require investment.

e. Priority level compared to other interoperability needs.

Assigning a moderate-to-high priority level is context-dependent. For chronic care, behavioral health, and multisite coordination, access to unstructured records is essential to effective care. In emergency or episodic care, structured data—such as allergies, medications, and lab results—remains foundational and should be prioritized first.

Strategically, CMS should prioritize the interoperability of structured data first. Concurrently, the agency should fund pilot programs and develop standards for scalable access to unstructured and semi-structured formats. To further enhance interoperability, CMS should incentivize vendors and providers to adopt AI-based data normalization tools as part of EHR certification and HIE participation.

PR-4. What changes or improvements to standards or policies might be needed for patients' third-party digital products to have access to administrative workflows, such as auto-populating intake forms, viewing provider information and schedules, and making and modifying an appointment?

We strongly support greater patient access to administrative workflows through third-party applications. Doing so improves patient experience, reduces front-office burden, and drives more equitable access to care. However, outdated policies, immature standards, and inconsistent vendor participation hinder progress on this front.

To empower patients and reduce friction in care access, CMS should expand the interoperability mandate beyond clinical data to cover administrative workflows. Standards must be clear, enforceable, and aligned with patient-facing app ecosystems. This will enable a more seamless, consumer-centric experience while also reducing burdens on clinical and administrative staff.

To ensure third-party digital products can seamlessly integrate into administrative workflows, several policy changes and improvements are necessary. First and foremost, it is critical to establish and enforce national standards for administrative APIs. Currently, while clinical data APIs such as FHIR US Core are evolving under ONC's 21st Century Cures Act rules, there is no equivalent mandate for administrative interoperability. Developing and adopting a standardized set of FHIR-based administrative APIs, such as FHIR scheduling, questionnaire, and insurance plan resources, would be a significant step forward. Furthermore, CMS and ONC should define and certify these APIs through EHR certification programs, specifically for appointment scheduling, provider directories, and intake forms.

Another essential improvement is expanding information blocking regulations to cover administrative data. At present, these rules focus primarily on clinical electronic health information (EHI). Consequently, patient-facing digital tools often encounter access denials for administrative tasks that are not explicitly covered by current regulations. CMS and ASTP/ONC need to clarify that administrative functions directly impacting access to care, such as scheduling and registration, fall

under anti-information blocking provisions. Additionally, incentivizing or mandating EHR vendors and health systems to make these administrative APIs accessible to third-party apps authorized by patients is crucial.

Improving access to provider directory and scheduling data is also necessary. Accurate and accessible provider availability and service-level data are often siloed or restricted to proprietary portals. Health systems and vendors should be required to expose real-time provider availability and appointment slots via standard APIs. Moreover, aligning these efforts with a National Directory of Healthcare Providers and Services (NDH) and future CMS interoperability rulemaking would ensure greater consistency and reliability.

Streamlining consent and authorization processes is vital. Currently, third-party apps face fragmented and manual consent processes when accessing appointment and intake functionalities. Promoting standardized OAuth2/OpenID Connect protocols for administrative access, similar to those required for patient data access under SMART on FHIR, would simplify these processes. Establishing granular, reusable patient consent standards for administrative data sharing with apps would further enhance efficiency and patient trust.

Providing external products with access to provider workflows raises significant privacy and security concerns. Without strict API governance, robust authentication (e.g., OAuth 2.0, FHIR scopes), and continuous monitoring, the attack surface expands. CMS could mandate a minimum-security certification or conformance testing process for third-party applications that access provider workflows.

Data Exchange

PR-6. Is TEFCA currently helping to advance provider access to health information?
a. Please provide specific examples.

TEFCA provides a clear framework for exchange purposes with a definition of Treatment, which is not clearly defined in other HIEs. Additionally, it is designed to include participants and qualified health information networks (QHINs) in the governance process to help shape the future of the network, which will help with participation. TEFCA has also encouraged standardization in identity verification and authorization models for cross-entity health data sharing. This has improved visibility and control over information flows, reducing certain cybersecurity risks.

b. What changes would you suggest?

The standard operating procedure for the Treatment exchange purpose must be updated to integrate non-covered entities while protecting patient data from non-clinical exchanges. For example, federally funded free clinics are not currently HIPAA-covered entities because there is no payment for their services. There should be an appropriate way to onboard free clinics without exposing TEFCA to all non-covered entities.

Additionally, there is a proposed use case for individual access services (IAS) to allow patients to self-direct their data. If TEFCA included this access, patient validation through identity verification,

and patient consent, more opportunities for data exchange beyond the Treatment use case would emerge.

TEFCA should also tackle the issue of a clear and concise provider database, including direct secure messaging addresses. And emphasizing TEFCA's "zero trust" principles, such as least privilege access, strong identity assurance, and continuous monitoring is essential to addressing cybersecurity concerns.

c. What other options are available outside of TEFCA?

While Care Everywhere is an option, its exchange capabilities are is limited to the Epic ecosystem. Carequality, Commonwell, and eHealth exchange are nationwide frameworks with which most organizations exchange, but because they are private, non-profit organizations, there are limited repercussions for bad actors who pull patient data claiming a Treatment relationship that is otherwise not valid. Because the data custodians that connect to these networks are still liable for the inappropriate disclosure of data, those exchanges are becoming riskier over time. TEFCA offers a federally supported exchange with ASTP/ONC oversight, which will allow clearer definitions and more appropriate consequences when an organization misrepresents itself.

d. Are there redundant standards, protocols or channels or both that could be consolidated?

We believe over time the best solution is to consolidate all HIEs to TEFCA. However, many health information organizations (HIOs), especially smaller ones, are hesitant to adopt the TEFCA framework due to financial concerns, legal uncertainties around privacy and compliance, and existing commitments to other data-sharing models. Technical challenges, such as interoperability and data integration, also pose barriers, though organizations using FHIR are more likely to participate. However, information might be limited since FHIR implementation is EHR vendor specific, leaving gaps between the FHIR standard and reality. Additionally, some QHINs, such as Epic's Nexus, are vendor-specific, limiting access for non-customers and forcing HIOs to join a different QHIN. As TEFCA participation is voluntary, adoptability remains a significant issue.

Participation in TEFCA and full implementation of the FHIR standard by EHR vendors should be mandatory. TEFCA should be the required method for exchanging information with government entities, and TEFCA should stand behind that requirement.

On the cybersecurity front, reducing fragmentation in HIEs could improve incident detection and response capabilities. A centralized TEFCA architecture enables better enforcement of uniform security controls and breach containment procedures.

PR-7. What strategies can CMS implement to support providers in making high-quality, timely, and comprehensive healthcare data available for interoperability in the digital product ecosystem? How can the burden of increasing data availability and sharing be mitigated for providers? Are there ways that workflows or metrics that providers are already motivated to optimize for that could be reused for, or combined with, efforts needed to support interoperability?

We recognize the critical importance of making healthcare data interoperable—both to meet regulatory goals and to enable digital innovation that improves care delivery, patient engagement, and

population health. To scale data availability without adding unsustainable burden to providers, CMS must align technical infrastructure, financial incentives, and operational workflows.

EHRs and other digital health vendors should be required to fully implement the FHIR standard and offer its use to HIOs with either no fees or a capped, affordable pricing structure to ensure equitable access and promote widespread interoperability.

Additionally, build between different health systems can lead to differences in information returned by FHIR API's which can have a significant impact on whether the APIs are useful for a particular implementation or whether an older standard like HL7 or a file extract needs to be used to get the information.

Security automation and standardized privacy policies can help providers meet data-sharing obligations without increasing burden. CMS could support this by aligning interoperability incentives with NIST-aligned cybersecurity practices and funding privacy-preserving technologies.

PR-8. What are ways CMS or partners can help with simplifying clinical quality data responsibilities of providers?
a. What would be the benefits and downsides of using Bulk FHIR data exports from EHRs to CMS to simplify clinical quality data submissions? Can CMS reduce the burden on providers by performing quality metrics calculations leveraging Bulk FHIR data exports?

We strongly support leveraging modern, scalable data exchange methods such as Bulk FHIR (Flat FHIR) to reduce fragmentation in clinical quality reporting. If executed thoughtfully, this approach can significantly reduce provider burden and increase data consistency, but it also introduces technical, governance, and policy considerations that must be addressed.

Bulk FHIR is a promising mechanism to modernize and streamline quality reporting. With the right policy, privacy, and infrastructure safeguards, CMS can reduce provider burden by centralizing quality measurement and harmonizing data pipelines. However, success will depend on investment in infrastructure, transparency in calculations, and continued efforts to close the gap between structured and unstructured clinical data.

Utilizing Bulk FHIR for quality data submissions offers several benefits. It significantly reduces redundant manual work by automating the extraction of relevant patient-level data in a standardized format. This automation diminishes the need for local mapping, data transformation, and custom file generation for each quality program.

Bulk FHIR leverages existing API infrastructure. Many certified EHRs are already mandated to support Bulk FHIR export capabilities under the ONC Cures Act Final Rule. This builds on the same ecosystem used for payer interoperability and patient access, enabling the infrastructure to serve multiple purposes.

Additionally, this approach enables CMS to perform centralized quality calculations, shifting the burden of measure computation and versioning to CMS or its designees. This ensures greater consistency in the application of measures, reducing variability across provider submissions. Moreover, Bulk FHIR supports real-time or near-real-time measurement. Scheduled Bulk FHIR exports, occurring periodically (e.g., daily or weekly), facilitate faster feedback loops and dynamic

performance dashboards. Furthermore, the reuse of Bulk FHIR extracts across programs amplifies the return on investment. The same data can be utilized for population health, value-based care contracts, and public health reporting, promoting multipurpose use.

Despite these advantages, there are downsides and challenges associated with the Bulk FHIR approach. Data completeness and fidelity concerns arise since Bulk FHIR currently focuses on structured data elements. Many quality measures depend on clinical nuances stored in unstructured data, scanned documents, or embedded workflows that may not be captured in Bulk FHIR resources.

Technical, cost, and infrastructure demands also present challenges. The implementation of FHIR is EHR vendor-specific and not fully standardized, with vendors charging for API access and usage, potentially making participation cost prohibitive. Generating large-scale, secure FHIR exports necessitates substantial compute, storage, and network bandwidth, particularly for multi-entity health systems. Small and rural providers may lack the resources to implement and maintain these capabilities without additional support.

Privacy, consent, cybersecurity, and governance issues are critical considerations. Patient-level data exports to CMS raise concerns about data minimization, patient consent, and secondary use, especially under HIPAA and other state privacy laws. Clear governance and use-case definitions must be established. Improperly secured Bulk FHIR exports pose serious risks, including large-scale PHI leakage. CMS should mandate encryption in transit and at rest, require access logging, and enforce identity verification for systems. Performing quality metrics calculations centrally can reduce exposure of sensitive data and ensure stronger, standardized protections.

Lastly, versioning and measure logic are essential for maintaining accuracy and fairness. If CMS assumes responsibility for quality calculations, it must be transparent, timely, and responsive to changes in measure specifications and clinical guidelines. Providers will still need the ability to preview and validate their performance locally to ensure that the calculations are fair and accurate.

b. In what ways can the interoperability and quality reporting responsibilities of providers be consolidated so investments can be dually purposed?

We see daily the strain that fragmented reporting and interoperability requirements place on our clinical, informatics, and IT teams. There is significant opportunity to consolidate quality reporting and interoperability efforts so that shared investments drive both regulatory compliance and care improvement. Below are key strategies CMS should consider:

- Aligning data standards across programs: Quality programs (e.g., MIPS, IQR, MSSP ACO) should be harmonized around a single FHIR-based data standard and shared core clinical data set (e.g., USCDI). This tactic allows providers to build once—capturing structured data at the point of care—and use it for both data exchange and quality measurement, thereby reducing duplicative abstraction and integration efforts. Using the same secure data pipeline (e.g., FHIR-based) for both interoperability and quality reporting can improve auditability and reduce redundant handling of sensitive data, enhancing security posture by minimizing data exposure.
- Creating a unified submission standard: Consolidating submissions to CMS through a single, digital gateway for both quality and interoperability programs (e.g., QPP, Promoting Interoperability Program) reduces the administrative overhead of managing multiple portals and submission protocols and promotes consistent data validation.

- Incentivizing the use of certified APIs that serve dual purposes: Requiring EHR vendors to expose certified APIs that support both clinical data exchange (e.g., via FHIR) and real-time quality measurement (e.g., DEQM) encourages investment in infrastructure that serves both compliance and care management, such as surfacing performance gaps within provider workflows.
- Promoting real-time quality feedback loops via interoperability rules: CMS should ensure that HIEs, registries, and payer data exchanges return clinically actionable, real-time quality performance data to providers. This would encourage dual use of interoperability data streams for care coordination and for tracking/reporting quality outcomes.
- Reducing redundancy across federal and state reporting programs: CMS should coordinate with state Medicaid programs and commercial payers to align quality measure definitions, submission timelines, and formats. Providers can leverage one data pipeline to serve multiple reporting obligations, increasing returns on interoperability investments.
- Funding and supporting shared infrastructure across health systems: CMS should expand support for regional data collaboratives or public-private HIEs that can centralize and normalize data for both interoperability and quality use cases. Smaller or rural providers can benefit from economies of scale, reducing the need for redundant point solutions.

By aligning quality measurement and interoperability requirements under common data, technical, and reporting frameworks, CMS can allow providers to make single investments that improve both compliance and clinical outcomes. This will accelerate the transition to a learning health system and reduce the burden on frontline teams already stretched by documentation and administrative tasks.

c. Are there requirements CMS should consider for data registries to support digital quality measurement in a more efficient manner? Are there requirements CMS should consider for data registries that would support access to real-time quality data for healthcare providers to inform clinical care in addition to simplifying reporting processes?

We fully support CMS's efforts to modernize quality measurement infrastructure. Current processes are fragmented, retrospective, and burdensome. Registries, if redesigned to support digital quality measurement (dQM) and real-time clinical utility, can serve as strategic assets rather than compliance repositories.

For registries to enable a future-ready quality ecosystem, CMS must set clear expectations for digital infrastructure, real-time feedback, and standards-based integration. These requirements will reduce clinician burden, improve data accuracy, and allow registries to become a proactive tool for care optimization, not just a passive reporting mechanism.

Requirements for Registries to Support Efficient Digital Quality Measurement

- Adopt Standardized Data Models and Exchange Formats: Registries should be required to adopt HL7 FHIR-based data models such as eCQMs, DEQM, and QI-Core to align with digital measure logic and modern health IT systems. This adoption would reduce custom extract/transmit logic and enable EHR vendors to support quality workflows natively within their platforms, streamlining the process and minimizing errors.
- Eliminate Manual Data Abstraction: Registries should disallow reliance on manual abstraction or PDF uploads, instead requiring automated ingestion via APIs or bulk FHIR data exchange. Manual processes are error-prone, delay measurement cycles, and significantly increase administrative burden.

- Ensure Registry Certification, Cybersecurity, and Governance: CMS should implement a certification program for digital-ready registries with clear criteria for interoperability, security, and timeliness. Registries should meet baseline cybersecurity controls aligned with NIST 800-53, offer real-time access logs, and enforce consent-based access. Additionally, registries should support secure APIs and conduct routine third-party risk assessments.

Requirements for Real-Time Quality Data Access to Support Clinical Care

- Near Real-Time Data Integration: Registries should support sub-daily data exchange, allowing providers to access updated quality metrics via standard APIs or dashboards. Real-time visibility into quality gaps at the point of care can improve delivery and reduce preventable harm, offering a proactive approach to patient care management.
- Bi-Directional Feedback and Benchmarking: Registries should support providers with the capability to both submit data and receive actionable insights, risk-adjusted performance benchmarks, and alerts on outlier performance. Empowering providers with comparative data facilitates targeted improvement efforts, fostering a culture of continuous quality improvement.
- Alignment with Clinical Workflows and Care Pathways: Registries should be designed to surface quality indicators within clinician-facing tools, such as EHRs and care management platforms, and allow patient-specific drilldowns. This transforms registries from retrospective tools into real-time quality intelligence platforms, seamlessly integrating with clinical workflows to enhance care delivery.

Simplifying Reporting While Enhancing Care

- Align Measurement with Care Improvement: CMS should promote "Digital First" Measures, encouraging measure developers to create digital-native eCQMs using standardized code sets and logic. Additionally, consolidating registry submissions to enable multi-program submission portals for clinical and claims-based registries would avoid duplication. Supporting provider self-service reporting by offering APIs that allow providers to run and preview quality measure performance internally before formal submission would further streamline the process and enhance accuracy.

Digital Identity

We are interested in standards that help us uniquely identify patients. A key interoperability challenge is mismatching patients with medical records. At best, the need to correct medical records is an administrative burden. At worst, patient misidentification has implications for patient safety and HIPAA violations. Preventable medical errors and safety issues as well as unauthorized disclosures can be avoided by accurately identifying patients and matching them with their records. We urge CMS to work with Congress on a national patient identifier strategy.

Information Blocking

PR-12. Should ASTP/ONC consider removing or revising any of the information blocking exceptions or conditions within the exceptions (45 CFR part 171, subparts B through D) to further the access, exchange, and use of electronic health information (EHI) and to promote market competition?

ASTP/ONC should consider information blocking requirements around administrative data, as discussed in our response to Question PR-4.

Meanwhile, the models for patient access via API should factor in that patient access to protected health information (PHI) should remain in line with clinical standards of care and safety to the extent possible. For example, delivery of test results and other health information should be within a model that can align with the clinical provider's review, accompanied with contextual information from the patient's provider, and then allow the patient to have an opportunity to receive provider counseling so the patient can understand the information and what the next steps may be from a care standpoint. As such, there may be scenarios where it could be detrimental to a patient and potentially not align with the standard of care to release PHI via an API without a more intentional and clinically driven workflow, such as confirmation of a diagnosis of cancer, prenatal genetic testing results, or information regarding rare degenerative conditions. It is also important that test results or other information are not delivered in real time while the patient may still be at their patient encounter, which could shift the burden of attempting to interpret the clinical information to the patient and could also cause a patient to leave without receiving the relevant clinical advice consistent with the standard of care. Therefore, there must be appropriate information blocking exceptions and/or flexibility to allow for clinical information to be released to patients in an appropriate manner and timeframe and in line with the standard of care for providers, while still prioritizing a patient's right to access their own health information.

PR-13. For any category of healthcare provider (as defined in 42 U.S.C. 300jj(3)), without a current information blocking disincentive established by CMS, what would be the most effective disincentive for that category of provider?

We urge the repeal the overly broad provider disincentives established in the rule entitled "21st Century Cures Act: Establishment of Disincentives for Health Care Providers That Have Committed Information Blocking" (89 Fed. Reg. 54662, RIN 0955–AA05). We recognize that steps should be taken to avoid harmful practices that obstruct the exchange of important health information and patient access to this information. Under this final rule, hospitals and providers found to engage in information blocking may face reductions in Medicare payment updates, adjustments to reimbursement rates, lower performance scores, and potential ineligibility for certain incentive programs. The disincentives are based on variable aspects of provider payment systems, such as the year-to-year changes in the value of the market basket adjustment and incentives given during different performance periods. This creates an unfair and confusing framework in which providers determined to be information blocking do not know what penalty to expect. Additionally, the final rule appears to create a system in which a hospital potentially could be penalized from multiple directions arising from a dispute over a single information access issue. The rule fails to clarify how HIPAA penalties and other existing accountability measures would interact with the disincentives and future penalties for providers.

Rather than imposing blanket penalties as finalized in this rule, a tiered enforcement framework would be a better approach. An appropriate enforcement process should begin with a warning, followed by education and technical assistance so that providers can understand what constitutes a violation, react to investigations and determinations, and take corrective measures. Such a process would set reasonable expectations for providers by following similar approaches in other compliance programs such as breach notifications and conditions of participation. Moreover, this framework would allow for agency discretion depending on whether a provider acted under a good faith interpretation of the relevant 21st Century Cures Act information blocking provisions or knowingly and intentionally committed information blocking.

VALUE-BASED CARE ORGANIZATIONS

Digital Health Adoption

VB-1. What incentives could encourage APMs such as accountable care organizations (ACOs) or participants in Medicare Shared Savings Program (MSSP) to leverage digital health management and care navigation products more often and more effectively with their patients?

It is crucial to prioritize technologies and vendors with a proven track record of engaging different types of consumers effectively. A user-friendly interface ensures that both providers and patients can navigate the digital tools with ease, thereby enhancing adoption and sustained usage. Furthermore, addressing consumer data privacy and security concerns is vital to build patient trust and compliance with digital health solutions.

Financial support plays a significant role in encouraging digital health adoption. Allocating funds specifically to APMs for the development and implementation of digital health tools can provide an initial boost. Additionally, offering advanced payments up front can help APMs invest in long-term digital strategies, ensuring they have the financial resources needed to integrate these technologies into their workflows.

Incentives for asynchronous or digital care management interactions, such as chatbots, secure messaging, and texting, without counting these expenditures in the APM entity's financial reconciliation, can further promote the use of digital tools. Additionally, ensuring interoperability and data standards that enable easy data transfer and seamless integration into existing EHRs in workflow-neutral ways will simplify the adoption process for APMs.

Enhancing transparency and timely data reporting to APMs regarding the reimbursement received by vendors with digital management platforms is crucial. This transparency ensures that APMs are fully aware of the financial implications and can make informed decisions regarding the adoption and utilization of these digital tools.

CMS should consider ways to support patients' access to wearable devices and digital apps to monitor and report health metrics for certain chronic conditions. Technology should also be leveraged to educate patients about their health. Targeted incentives, such as premium discounts or cost sharing waivers, can encourage patient engagement with their health, including through digital platforms, ultimately contributing to enhanced continuous care, better health outcomes, and savings. For example, through Cleveland Clinic's employee health plan, we have found that 74% of participants either report engagement in a defined list of healthy behaviors or participate in care management through our Healthy Choice digital platform in order to obtain a discount on their insurance premiums.

Allowing digital platforms, such as patient-reported tools and home monitoring of vitals, to capture clinical data to contribute to risk adjustment and Hierarchical Condition Category (HCC) coding will further incentivize adoption by integrating these new data sources into existing healthcare assessment frameworks.

By addressing these key areas with targeted incentives, APMs can be encouraged to leverage digital health management and care navigation products more effectively, thereby enhancing patient care and operational efficiency.

What are the current obstacles preventing broader digital product adoption for patients in ACOs?

ACOs face numerous challenges that hinder the adoption of digital products for their patients. One significant obstacle is the complexity of model design and conflicting incentives. These challenges leave ACOs with limited bandwidth to develop and implement digital tools effectively.

Financial constraints also play a crucial role. Investing in the expertise needed to develop digital tools is unaffordable for most ACOs. Additionally, the variable nature of ACO revenues makes long-term investment in digital tools less of a priority compared to other infrastructure investments, such as care management.

Another barrier is the disparity in digital literacy and access among ACO participants. Elderly patients and those in rural areas with poor broadband access struggle to engage with digital solutions. This disparity further complicates the adoption of digital products.

Furthermore, the ease of integration of digital tools and patient-reported information into EHRs is a critical concern. Fragmentation across vendors and tools creates challenges for patients, healthcare workers, and analysts, leading to difficulties in seamless integration.

Regulatory and compliance requirements add another layer of complexity, leaving ACOs with little incentive to invest in digital tools. Combined with a lack of trust and skepticism towards these technologies, ACOs may find themselves disinclined to adopt digital products.

VB-2. How can key themes and technologies such as artificial intelligence, population health analytics, risk stratification, care coordination, usability, quality measurement, and patient engagement be better integrated into APM requirements?

Developing an APM model that offers upfront investments to ACOs can empower them to create a toolkit of evidence-based capabilities. This toolkit should include tools for risk stratification, care coordination, quality measurement, and patient engagement. By providing guidance and benchmarks for these tools, CMS can ensure that ACOs are adequately equipped to enhance patient care and operational efficiency.

Moreover, CMS can measure patient engagement as an outcome and reward higher levels of engagement with increased per member per month payments or bonuses. This would incentivize ACOs to adopt digital tools that promote patient involvement and improve health outcomes.

Additionally, a sliding scale for investment can be developed by CMS, allocating more funds to geographies or ACOs that require greater encouragement for member participation. This targeted approach could provide ACOs with resources to address disparities in digital literacy and access, particularly among elderly patients and those in rural areas with poor broadband connectivity.

Finally, CMS should publicly endorse a framework that provides incentives for the adoption and interoperability of digital tools. By actively promoting the integration of these technologies, CMS can foster a more cohesive and efficient healthcare system that benefits both patients and providers.

VB-3. What are essential health IT capabilities for value-based care arrangements? What other health IT capabilities have proven valuable to succeeding in value-based care arrangements?

The ability to manage member rosters and stratify population risk is essential in value-based arrangements. By understanding the demographics and risk profiles of the population, healthcare organizations can more effectively tailor their interventions.

Quality measurement is another critical capability, which involves aggregating data from clinical records, claims, and public health sources to measure outcomes. This ensures that care management platforms and patient wellness apps are effectively engaging patients and improving their health.

Documentation is crucial for adjusting population risk. Accurate and comprehensive documentation helps to ensure that the risk profiles of patients are well understood and managed.

Utilization measurement is essential for tracking high-cost, high-impact healthcare services such as admissions, emergency department visits, skilled nursing facility stays, and high-cost drugs and procedures. This measurement relies on data aggregated from clinical, claims and public health sources. Additionally, patient access and scheduling platforms, along with telehealth apps and care cost calculators, play a vital role in making healthcare services more accessible and affordable.

Lastly, network management capabilities are indispensable. A provider referral platform, along with tools for network profiling and benchmarking, ensures that healthcare providers can collaborate effectively and maintain high standards of care.

VB-4. What are the essential data types needed for successful participation in value-based care arrangements?

To successfully participate in value-based care arrangements, healthcare organizations must leverage several essential data types across multiple domains. To manage member attribution, they need detailed demographic information that helps identify and understand the population being served. Quality measurement demands comprehensive clinical data, including patient notes, lab results, imaging studies, patient experience metrics, and engagement data. Integrating patient-reported and patient-submitted information also plays a crucial role in evaluating the effectiveness of care.

Accurate documentation of clinical data is vital for adjusting population risk profiles. This practice ensures that the risk associated with each patient is well-documented and managed effectively. Utilization measurement necessitates the aggregation of clinical data, claims records, public benchmarks, cost of care information, and insights into non-clinical barriers driving utilization, such as social needs. Predictive analytics are also employed to anticipate and manage healthcare needs proactively.

Finally, network management relies on provider panel and outcomes analytics to ensure that healthcare providers can collaborate efficiently and maintain high standards of care. These data help

in profiling and benchmarking networks, enabling healthcare organizations to deliver high-quality, efficient, and patient-centered care.

Compliance and Certification

VB-7. How can technology requirements for APMs, established through CEHRT or other pathways, reduce complexity while preserving necessary flexibility?

Technology requirements for APMs established through Certified Electronic Health Record Technology (CEHRT) or other pathways can simplify processes while maintaining essential flexibility. By defining a small set of non-negotiable capabilities that must be standard across systems, we can achieve both standardization and clarity. This approach promotes the development of plug-and-play ecosystems, making it easier to integrate new capabilities. Additionally, it standardizes and incentivizes connectivity for national health information exchange, ensuring seamless and efficient data sharing.

VB-8. How can other HHS policies supplement CEHRT requirements to better optimize the use of digital health products in APMs? As an example, requirements under the Conditions of Participation for hospitals (42 CFR 482.24(d)) require hospitals to transmit electronic patient event notifications to community providers. What barriers are in place preventing APM participants from receiving the same notifications?

Mandating admission, discharge, and transfer (ADT) event notification and real-time alert connectivity between all contracted providers and EHRs could incentivize HIE subscriptions. However, several barriers exist, including the ability of patients to opt-out of event notification sharing, integration costs pertaining to time, money, and capacity constraints, a lack of incentives for data sharing, and geographic or EHR-based silos around HIEs.

VB-9. What technology requirements should be different for APM organizations when comparing to non-APM organizations (for example, quality reporting, and interoperability)?

Any requirement should recognize that technology needs for APM organizations differ from those of non-APM organizations in several critical ways. For one, quality reporting should be more comprehensive for APMs, encompassing not only traditional metrics but also health disparity reporting to ensure equity in care delivery. Interoperability becomes paramount, as seamless data exchange between different systems can significantly enhance patient outcomes and operational efficiencies. Telehealth capabilities should be robust and integrated, facilitating access and continuity of care across diverse geographic locations.

Additionally, APM organizations benefit from the implementation of cost-of-care calculators, providing both providers and patients with transparency and informed decision-making tools. Claims aggregation is another essential requirement, enabling the consolidation of billing information to streamline administrative processes. Secure communication channels between EHRs are vital, promoting confidentiality and accuracy in patient data exchanges.

Risk stratification, supported by clinical decision support tools, allows for better identification and management of high-risk patients, ensuring that preventative measures are effectively targeted. Shared care plans and patient goals foster collaboration between healthcare providers and patients,

enhancing engagement and adherence to treatment plans. Finally, care gap notifications alert providers to missed or overdue care interventions, contributing to improved patient care and outcomes.

In essence, the distinct technology requirements for APM organizations reflect a deeper commitment to comprehensive, coordinated, and patient-centric care, driving advancements in healthcare delivery and outcomes.

VB-10. In the Calendar Year (CY) 2024 Physician Fee Schedule final rule (88 FR 79413), CMS established that CEHRT requirements for Advanced APMs beyond those in the "Base EHR" definition should be flexible based on what is applicable to the APM that year based on the area of clinical practice. What certification criteria should CMS identify under this flexibility for specific Advanced APMs, or for Advanced APMs in general? Are there specific flexibilities or alternatives to consider for smaller or resource-constrained (such as rural) providers in meeting CEHRT requirements without compromising quality of care or availability of performance data?

The final rule ensured that all MSSP participants within an ACO are meeting the established CEHRT requirements by 2025. This rule reinforced that ACOs add these criteria in evaluating current and future participants. These requirements should be maintained to ensure consistency of reporting capabilities across participants within an ACO.

Technical Standards

VB-11. What specific interoperability challenges have you encountered in implementing value-based care programs?

One of the major challenges with implementing value-based care programs is the inability to seamlessly exchange data across different healthcare organizations, states, or multiple payers, or with CMS, largely due to technology constraints, regulatory requirements, lack of standardization, and the cost of implementation. Interoperability necessitates significant administrative investment to implement and manage, including the need for a specialized workforce and potentially the engagement of external aggregator vendors.

To illustrate, real-time ADT notifications can support prompt and efficient data sharing, yet their integration is often hampered by technological and financial barriers. Moreover, there is a notable lack of integration with community-based organizations, which is crucial for incorporating social services that significantly impact patient care.

Additional challenges include the requirement for real-time administrative investment to support interoperability efforts, and the complexity of coordinating care among diverse providers, which can lead to fragmented information and hinder comprehensive patient care.

VB-13. What improvements to existing criteria and standards would better support value-based care capabilities while reducing provider burden?

To better support value-based care capabilities while reducing provider burden, it is essential to establish a standardized format akin to the 837 electronic file claim transaction used in fee-for-service models, but specifically tailored to value-based care. This format should be integrated into a FHIR API to ensure seamless data exchange. Standardization would encompass claim elements and

eligibility criteria, with specific field names, field types, and data consistently defined. Moreover, incentivizing outcome-based performance would further enhance the efficiency and effectiveness of value-based care programs. By addressing these criteria and standards, the healthcare system can streamline processes, reduce administrative costs, and ultimately improve patient outcomes.

VB-14. How could implementing digital identity credentials improve value-based care delivery and outcomes?

Introducing digital identity credentials can significantly enhance value-based care delivery and outcomes by reducing error rates in member identification and maintaining the integrity of health records across various care providers. This system facilitates the tracking of individual and population health trends over time and has the potential to streamline plan enrollment processes. Moreover, it improves coordination among care team members and strengthens the security of EHRs, ensuring more timely and efficient access to necessary information. Additionally, digital identity credentials enhance connections with community-based organizations, enabling the sharing of data on social needs.

VB-15. How could a nationwide provider directory of FHIR endpoints help improve access to patient data and understanding of claims data sources? What key data elements would be necessary in a nationwide FHIR endpoints directory to maximize its effectiveness?

A nationwide provider directory of FHIR endpoints would facilitate seamless data exchange across various healthcare providers, without the need for the cumbersome process of downloading, transferring, and uploading files, thus improving continuity of care. Such a directory should include eligibility dates for members and demographics because the foundation of data exchange begins with eligibility verification. Additionally, the directory should provide a comprehensive history of medical and prescription claims for members, even prior to their eligibility start dates, to ensure healthcare providers have a complete understanding of the patient's medical history. This includes revenue codes, CPTs, and Rx NDCs for each encounter, supporting thorough and accurate claims data analysis.

Thank you for conducting a thoughtful process that allows us to provide input on such important issues and for your consideration of this information. Should you need any further information, please contact me at myersm21@ccf.org.

Sincerely,

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